

Customization of 2D Atomic-Molecular Heterojunction with Manipulatable Charge-Transfer and Band Structure

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Manipulating the properties of 2D materials through meticulously engineered artificial heterojunctions holds great promise for novel device applications. However, existing research on the crucial charge-transfer interactions and energy profile regulation is predominantly focused on 2D van der Waals structures formed via weak van der Waals forces, limiting regulatory efficiency at high costs. Herein, a refined atomic-molecular heterojunction strategy featuring strong covalent bonds between organic molecule and 2D violet phosphorus (VP) atomic crystal is developed, which enables enhanced charge-transfer dynamics and customizable band structure regulation at the molecular level. Both experimentally and theoretically, it is demonstrated that grafting efficiency, charge redistribution, and energy gap regulation critically depend on organic electronegativity, providing a low-cost yet high-efficiency regulatory effect on a large scale. As a proof of concept, the novel VP-molecular heterojunctions exhibit optimized performance in diverse application domains, presenting a general platform for future high-performance device applications.

1. Introduction

2D materials, distinguished by their atomically flat surfaces and exotic quantum confinement effects, hold immense promise for the development of innovative 2D heterojunctions.^[1–3] These unique artificial configurations are achieved by vertically integrating 2D materials with a diverse range of material species, including analogous 2D atomic crystals,^[4,5] bulk semiconductor films,^[6,7] perovskites,^[8,9] and organic molecules,^[10,11] etc. By

capitalizing on the synergistic advantages of individual components, as well as the favorable charge-transfer interactions,^[12,13] bandgap tuning strategies,^[14] and defect passivation effects,^[15] 2D heterojunctions have made significant advancements in the fields of high-performance transistors,^[4,16] photodetectors,^[17,18] and photovoltaic cells.^[19,20]

In general, 2D heterojunctions are categorized into non-covalent (van der Waals) heterojunctions and covalent heterojunctions based on the bonding characteristics of the heterogeneous species.^[21] Non-covalent heterojunctions, typified by conventional 2D atomic crystal heterojunctions and organic-inorganic hybrid heterojunctions, rely on weak van der Waals forces due to the van der Waals gap between the disparate constituents, constituting the predominant paradigm.^[22] However, the intricate fabrication procedures

and limited overall regulation efficiency currently impede their widespread applications. As a competitive paradigm, 2D atomic-molecular heterojunctions, formed by grafting guest organic molecules onto the surface of host 2D materials via covalent bonds, demonstrate exceptional performance advantages for large-scale, customizable, and cost-effective preparation in practical applications. In contrast to noncovalent counterparts, robust covalent bonds rather than van der Waals gaps exist between organic molecules and 2D materials. The fundamentally enhanced charge-transfer interactions and degrees of charge redistribution thereby further improve device performance and operational stability. To date, there have been many reports on optimizing device performance through covalent modification, e.g., passivating defects,^[15] regulating carrier doping types,^[23] and reducing contact resistance.^[24] However, either limited by the naturally inert surface of 2D materials leading to low modification efficiency,^[25,26] or constrained by the limited range of organic molecules involved,^[27,28] it remains a daunting challenge to systematically investigate the fundamental regulatory principles that govern the interaction of specific organic molecules with 2D materials.

As another allotrope of the phosphorus family, 2D violet phosphorus (VP) exhibits superior thermodynamic and ambient stabilities compared to black phosphorus (BP),^[29,30] along with outstanding electrical and photoresponse characteristics.^[5,31,32] Unlike transition metal dichalcogenides (TMD), VP features a

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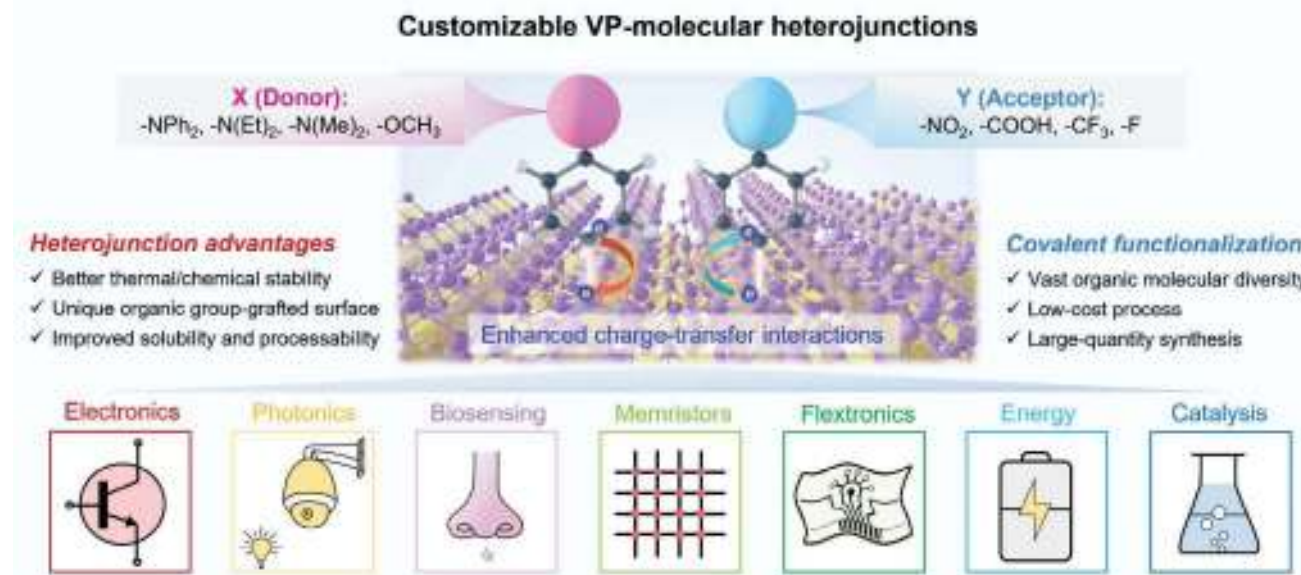


Figure 1. Schematic illustration depicting customizable VP-molecular heterojunctions with enhanced charge-transfer interactions and their potential applications. X (Donor) represents triphenylamine, N,N-Diethylaniline, N,N-Dimethylaniline, and methoxybenzene, respectively. Y (Acceptor) represents nitrobenzene, benzoic acid, trifluorotoluene, and fluorobenzene, respectively.

surface abundant with intrinsic lone-pair electrons, offering numerous reactive sites that enhance covalent modification efficiency. In this work, we synthesized eight VP-molecular heterojunctions and systematically investigated the differences in modification degree, charge transfer, and band structure regulations induced by organic molecules with varying electronegativities. Our findings, from both experimental and theoretical perspectives, indicate that organic molecules efficiently graft onto the surface of VP, significantly modulating its band structure and charge-transfer characteristics. Due to differences in electronegativity, both modification efficiency and band structure regulation exhibit variability. Specifically, organic molecules with electron-accepting properties demonstrate significantly higher modification efficiency compared to their electron-donating counterparts due to higher formation energy. Electron-donating molecules lead to a greater reduction in the bandgap (E_g) of VP relative to electron-accepting counterparts. By leveraging enhanced charge-transfer interactions and differential band structures, we demonstrated the potential of these novel atomic-molecular heterojunctions in applications such as photodetectors, humidity sensors, and memristors, resulting in significant performance improvements.

2. Results and Discussion

2.1. Design and Preparation of Customizable VP-Molecular Heterojunctions

To prepare high-quality VP-molecular heterojunctions with varying organic electronegativities, the primary objective is to establish an organic molecule system exhibiting distinct electron-donating and electron-accepting properties. In this regard, we carefully selected eight organic molecules—four with electron-donating and four with electron-accepting properties—that dis-

play a range of electronegativities (Note S1, Supporting Information). Few-layer VP nanosheets (VPNSs) were obtained through standard liquid-phase exfoliation coupled with lithium intercalation, which reduced the sample thickness and improved subsequent covalent functionalization efficiency.^[33,34] The aryl diazonium functionalization method, widely used in the functionalization of 2D materials,^[35,36] was employed to prepare VP-molecular heterojunctions with different substituents in the para position (see Experimental Section and Note S2, Supporting Information). Four VP-Donor heterojunctions—VP-TPA, VP-N(Et)₂, VP-N(Me)₂, and VP-OCH₃—were synthesized by functionalizing VP with electron-donating molecules: triphenylamine (TPA), N,N-Diethylaniline, N,N-Dimethylaniline, and methoxybenzene, respectively. In contrast, four VP-Acceptor heterojunctions—VP-NO₂, VP-COOH, VP-CF₃, and VP-F—were synthesized by incorporating electron-accepting organic molecules: nitrobenzene, benzoic acid, trifluorotoluene, and fluorobenzene, respectively.

As schematically illustrated in Figure 1, the robust C-P covalent bonds between VP and the organic molecules impart superior thermal and chemical stabilities to the synthesized VP-molecular heterojunctions compared to pristine VP and van der Waals counterparts. The extensive diversity of organic molecules, coupled with the cost-effective, large-scale synthesis process, endows these novel atomic-molecular heterojunctions with unique organic group-grafted surfaces and enhanced processability, making them highly promising for practical applications. Moreover, the reconfigured dipolar interactions between the covalently bonded organic molecules and the host VP induce enhanced charge-transfer interactions, which can be precisely tuned by selecting different organic donors or acceptors. As a result, the customizable VP-molecular heterojunctions can manipulate the carrier transport dynamics and electronic band structures at the molecular level, exhibiting great application potential in various domains, e.g., high-performance electronics,

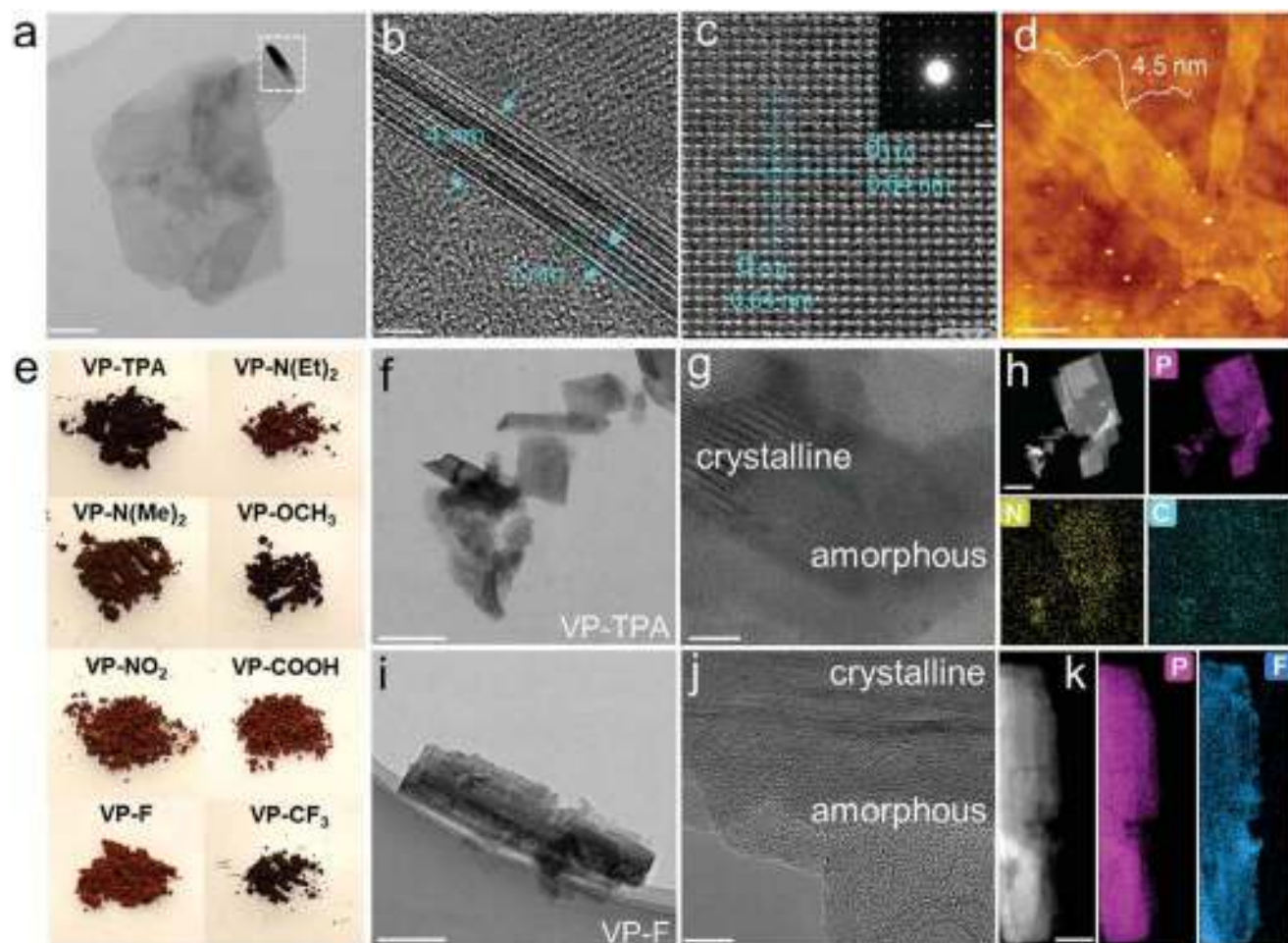


Figure 2. Structural characterization of intrinsic VP and VP-molecular heterojunctions. a) TEM image of a typical VP nanosheet. Scale bar: 50 nm. b) HRTEM image showing the upturned edge of the sample, indicated by the white square in (a). Scale bar: 2 nm. c) HRTEM image of VP nanosheet. Scale bar: 2 nm. Inset: the corresponding SAED pattern of the same area. Scale bar: 1/nm. d) AFM image showcasing a VP nanosheet on SiO₂/Si substrate. Scale bar: 3 μ m. e) Optical images featuring eight synthesized VP-molecular heterojunction powders. f) TEM image of the VP-TPA heterojunction nanosheets. Scale bar: 100 nm. g) HRTEM image revealing different crystalline properties within distinct areas of the VP-TPA heterojunction nanosheet. Scale bar: 200 nm. h) STEM image of the VP-TPA heterojunction nanosheets along with corresponding EDX mapping of P, N, and C elements. Scale bar: 200 nm. i, j) TEM and HRTEM images of the VP-F heterojunction nanosheet, respectively. Scale bars: 200 nm and 10 nm. k) STEM image of the VP-F heterojunction nanosheet with corresponding EDX mapping of P and F elements. Scale bar: 200 nm.

photonics, biosensing devices, memristors, energy storage/conversion, and catalysis, etc.

2.2. Structural Characterization and Bandgap Regulation

After the exfoliation and chemical synthesis processes, the morphology and structural characteristics of the prepared VPNSs and VP-molecular heterojunctions were investigated. Figure S3 (Supporting Information) clearly shows dispersive VPNSs on the SiO₂/Si substrate with almost transparent, leaf-like characteristics, indicating a uniform exfoliation effect. Figure 2a–c display the transmission electron microscopy (TEM) and high-resolution TEM (HRTEM) images of a typical VP nanosheet. The pristine sample shows a neat and smooth surface, with an upturned edge indicating a thickness of $\approx$ 4 nm. The HRTEM image, coupled with the corresponding sharp and square-like selected area elec-

tron diffraction (SAED) pattern, provides strong evidence that the VPNSs maintain an excellent crystalline structure after exfoliation. Atomic force microscopy (AFM) measurement in Figure 2d reveals that the average thickness of VPNSs is $\approx$ 4.5 nm, consistent with the TEM result, demonstrating that the exfoliated VPNSs are approximately four layers thick. The obtained high-quality and ultrathin VPNSs are essential for the sequent high-efficiency organic functionalization.

Following the diazo salt modification process, the optical images of the eight synthesized VP-molecular heterojunction powders, shown in Figure 2e, exhibit distinct colors, confirming the diverse functionalization effects of the various organic molecules. TEM characterization was performed to investigate the morphological differences after functionalization. As illustrated in Figure 2f, several flocculent structures emerge upon the surface of the VP-TPA sample compared to the pristine counterpart, leading to an aggregating state. HRTEM was

employed to elucidate the specific structure of VP-TPA. As shown in Figure 2g, two distinct areas, crystalline and amorphous, are observed on the same VP-TPA nanosheet. The crystalline region, characterized by clear lattice fringes, represents the main body of the VP sample, while the amorphous characteristic likely arises from loosely bonded TPA molecules on the surface, significantly affecting the diffraction pattern. Additionally, scanning transmission electron microscopy-energy dispersive X-ray spectroscopic (STEM-EDX) measurement was performed to intuitively map the chemical distribution within VP-TPA. Figure 2h exhibits that P, N, and C elements are homogeneously distributed over a typical VP-TPA nanosheet, with the N and C elements originating from the modified TPA molecules, indicating a uniform functionalization effect. Similarly, Figure 2i–k demonstrates the structural and compositional characterizations of the VP-F nanosheet, showing comparable results. Additional detailed TEM and STEM analyses of the remaining six samples, provided in Note S3 (Supporting Information), further confirm similar morphological and compositional characteristics across the different VP-molecular heterojunctions. These characterizations collectively indicate that the synthesized VP-molecule heterojunctions not only retain the crystalline structure of the intrinsic VP but are also homogeneously coated with the different organic molecules, showcasing the significant potential for further performance regulation.

It is intriguing to explore how the electronegativity of organic molecules affects the surface chemistry and E_g regulation of VP-molecule heterojunctions. To this end, a series of spectroscopic and quantitative compositional characterizations were conducted. Micro-Fourier transform infrared spectroscopy (Micro-FTIR) was employed to substantiate the covalent functionalization nature of the prepared VP-molecule heterojunctions. Figure 3a compares the Micro-FTIR spectra of the pristine bulk VP, VP-TPA, and VP-F. Since pristine VP is non-polar and lacks a dipole moment, no distinct vibrational peaks are observed except for a broad peak $\approx 1640\text{ cm}^{-1}$, corresponding to environmental water. In contrast, both VP-TPA and VP-F exhibit multiple prominent vibrational peaks, which can be categorized into C–P bond vibrations and specific functional group vibrations. Specifically, the new peaks at 1450 and 805 cm^{-1} in VP-TPA, as well as 1454 and 798 cm^{-1} in VP-F, are attributed to the C–P bond vibrations, providing solid evidence that the guest organic molecules are covalently bonded onto the host VP.^[34,37] Additionally, the sharp C–N stretching at 1310 cm^{-1} and C–F stretching at 1157 cm^{-1} correspond to the intramolecular vibrations of TPA and fluorobenzene, respectively.^[38,39] More detailed characterizations of all eight VP-molecule heterojunctions are presented in Note S4 (Supporting Information). In all cases, the observed shifts in C–P bond vibrations from 795 to 852 cm^{-1} , alongside distinct specific functional group vibrations, highlight the corresponding bonding environment changes and generalized functionalization effects. Moreover, Raman spectroscopy also corroborates the covalent functionalization. As illustrated in Figure 3b, the Raman spectra of eight VP-molecule heterojunctions exhibit shifts of $\approx 1\text{--}3\text{ cm}^{-1}$ along with reduced intensities compared to the pristine VPNSs. These alterations indicate changes in the P atom vibrational environment, likely resulting from the steric hindrance imposed by the covalently attached organic molecules and intermolecular interactions within the heterojunctions.^[28,40]

To further quantify the degree of covalent functionalization in the VP-molecule heterojunctions, elemental analysis was conducted. The molar ratios of host VP to guest organic molecules, determined by the P/C mass ratios, were used to assess the grafting efficiency. As depicted in Figure 3c and detailed in Note S5 (Supporting Information), all eight samples demonstrate significantly higher degrees of functionalization compared to TMDs counterparts,^[41,42] and even surpass that of similar BP heterojunctions.^[27,28] Notably, the grafting efficiency correlates with the electronegativity of the organic molecules. The four VP-Acceptor heterojunctions demonstrate significantly higher grafting efficiencies than the VP-Donor heterojunctions, with the average efficiency being ≈ 3 times greater. This correlation between organic electronegativity and grafting efficiency paves a pioneering path for precisely regulating the comprehensive properties of atomic-molecular heterojunctions. To elucidate the underlying mechanisms, we also conducted density functional theory (DFT) calculations, which will be discussed in the subsequent section. The maximum grafting efficiency exceeds 3.5% for VP-NO₂ heterojunction, underscoring a notably high functionalization level. Specifically, the optimized atomic configurations of such VP-NO₂ heterojunction, determined by DFT calculations (see Experimental Section), are visualized from both top and bottom views (upper panels of Figure 3d,e). The projections of the nitrobenzene clusters are then illustrated to assess the comprehensive surface coverage rate of the tethered organic molecules upon the VP nanosheet. Due to the non-negligible steric hindrance between adjacent organic pendants, the surface arrangements of the individual nitrobenzene molecules vary, leading to diverse projection areas on the VP nanosheet (lower panels of Figure 3d,e). The overall surface coverage rate of the VP-NO₂ heterojunction, calculated by summing the projected area ratios of nitrobenzene molecules from both upper and lower views of the host VP, reaches 13.99% , underscoring a distinguished covalent functionalization effect. At the material level, such high-quality covalent functionalization also contributes to enhancing the stability of VP-molecular heterojunctions, underscoring exceptional protective effects in practical applications.

Photoluminescence (PL) and ultraviolet–visible–near infrared absorption (UV–Vis–NIR) spectra were employed to elucidate the excitonic recombination process and optical E_g of the eight synthesized VP-molecule heterojunctions. As shown in Figure 3f, the PL peak of the pristine VPNSs is centered at 662 nm , consistent with previous reports.^[43,44] After covalent functionalization, significant redshifts in the PL peaks are observed for both VP-Donor and VP-Acceptor heterojunctions. These redshifts are attributed to radiative recombination at shallow surface states introduced by the modifications.^[45] Interestingly, VP-Donor heterojunctions exhibit a broader redshift range ($\approx 44\text{ nm}$) compared to the VP-Acceptor heterojunctions ($\approx 28\text{ nm}$), indicating that more shallow energy levels were introduced within the bandgap of VP-Donor heterojunctions, thus contributing to a narrower E_g . The variations in PL peak positions and corresponding optical E_g values of pristine VP and VP-molecule heterojunctions, as determined by UV–Vis–NIR spectroscopy, are depicted in Figure 3g, showcasing a consistent trend. Donor molecules generally induce a more substantial reduction in optical E_g than acceptors, with the maximum reduction rate reaching up to 25% for VP-TPA, highlighting the potential for extensive band structure

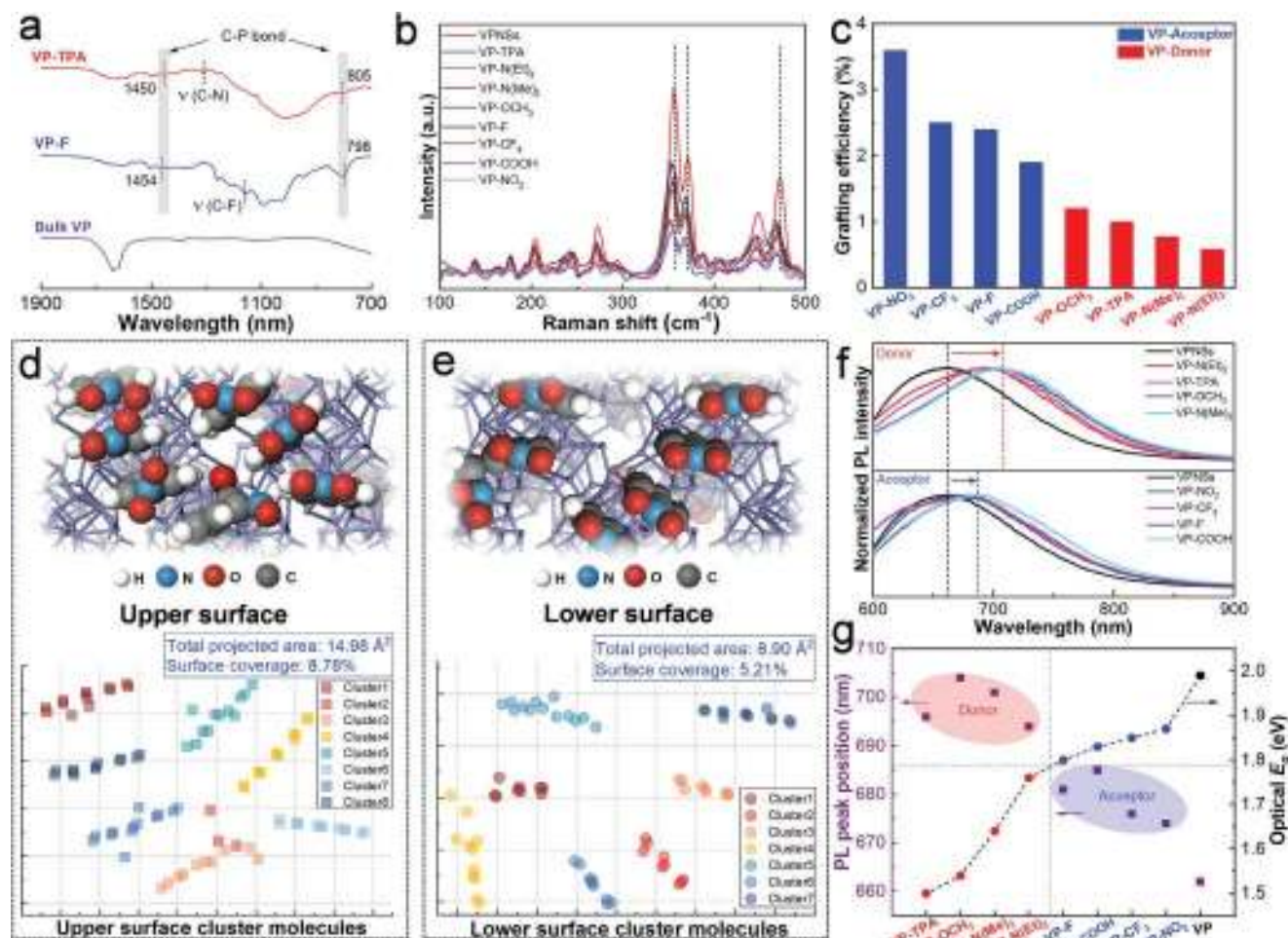


Figure 3. Spectral characterization, atomic configuration, and grafting efficiency quantification of VP-molecular heterojunctions. a) FTIR spectra of the bulk VP (violet), VP-TPA (red) and VP-F (blue). b) Raman spectra comparing pristine VP with eight synthesized VP-molecular heterojunctions. c) Comparison of quantified grafting efficiency between VP-Acceptor and VP-Donor heterojunctions. d,e) Optimized atomic configurations of VP-NO₂ heterojunction with 3.5% grafting efficiency, and projection of the tethered nitrobenzene molecules onto the 2D VP nanosheet, from top (d) and bottom (e) views, respectively. Note that the black orthogonal frames in the lower panels represent the total areas of the 2D VP nanosheet, while the colored square and circular spots stand for the projected individual clusters of different nitrobenzene molecules grafted onto the VP nanosheet. Due to molecular deformation poststructure optimization, the same organic molecule may appear as separate clusters; for example, cluster1 and cluster2 on the upper surface belong to the same nitrobenzene molecule. f) PL spectra shift of VP-Donor and VP-Acceptor heterojunctions compared with pristine VP. g) Trends in PL peak positions and optical E_g variations of VP and VP-molecular heterojunctions.

regulation. Again, considering the current covalent efficiencies, it is anticipated that further increasing the grafting rate of organic molecules could yield even more significant E_g modulation, meriting further detailed investigation in future studies.

2.3. DFT Calculations on Electronic State, Band Structure, and Reaction Dynamics of VP-Molecular Heterojunctions

The above-verified covalently grafted organic molecules, each possessing distinct electronegativities, exert diverse impacts on the charge-transfer interactions and electronic band structures of VP-molecule heterojunctions. To explore these regulatory effects, DFT calculations were conducted. Initially, the charge density difference ($\Delta\rho$) was computed, defined as $\Delta\rho = \rho(\text{VP-molecular heterojunctions}) - \rho(\text{VP}) - \rho(\text{organic molecule})$, where

$\rho(\text{VP-molecular heterojunction})$, $\rho(\text{VP})$ and $\rho(\text{organic molecule})$ represent the charge densities of the heterojunction, the pristine VP, and the organic molecule, respectively. This calculation enables the visualization of electronegativity-dependent charge-transfer interactions, providing critical insights into how different molecules modulate the electronic properties of the heterojunctions. As depicted in Figure 4a–e, donor molecules such as TPA and N,N-Diethylaniline facilitate a noticeable charge transfer from the guest molecules to the VP, enhancing electron density on the VP surface. Conversely, acceptor molecules such as nitrobenzene and benzoic acid induce an opposite effect, prompting charge transfer from the VP to the guest molecules, and reducing electron density in VP. Moreover, the electronic band structures of four VP-molecular heterojunctions are exhibited in Figure 4f–i, revealing calculated E_g values for VP-TPA, VP-N(Et)₂, VP-COOH, and VP-NO₂ as 0.76, 1.30, 1.40, and 1.75 eV,

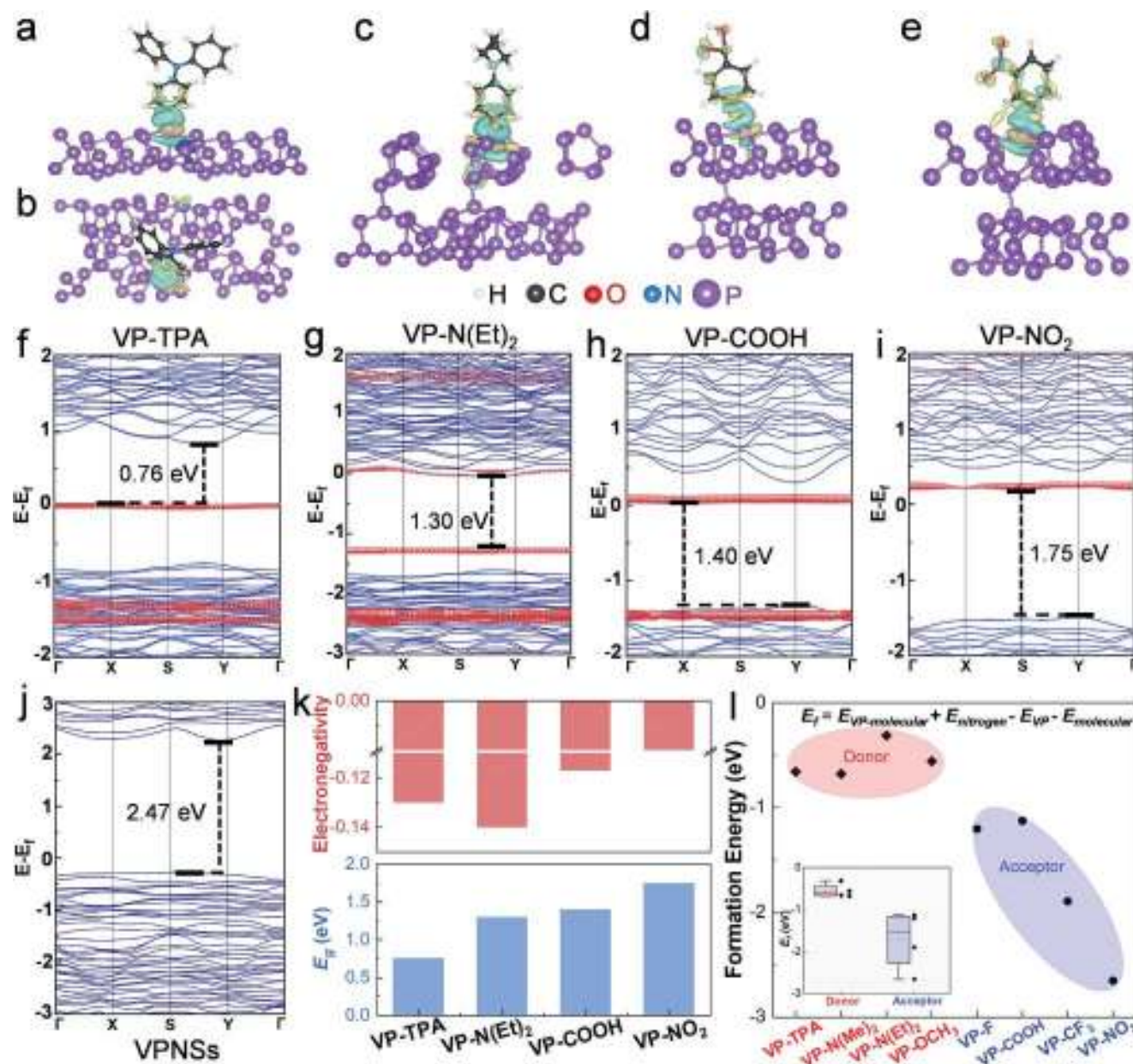


Figure 4. DFT calculations about differences in charge density, band structure, and formation energy of VP-molecular heterojunctions. a,b) Schematic diagrams of side and top views of charge density differences of VP-TPA, respectively. c–e) Schematic diagrams of side views of charge density differences of VP-N(Et)₂ c), VP-NO₂ d), and VP-COOH e) heterojunctions, respectively. In all cases, cyan and yellow colors represent electron depletion and accumulation, respectively. Isosurface: 0.002 e Å⁻³. f–j) Band structures of VP-TPA f), VP-N(Et)₂ g), VP-COOH h), VP-NO₂ i) heterojunctions, and pristine VP j), respectively. k) Variation of E_g regarding organic electronegativity of VP-TPA, VP-N(Et)₂, VP-COOH and VP-NO₂. l) Comparison of formation energy between VP-Donor and VP-Acceptor heterojunctions. Inset: Corresponding distribution comparison of formation energy.

respectively. Compared to the intrinsic E_g of the VP nanosheet in Figure 4j, all VP-molecular heterojunctions exhibit E_g reductions, attributable to the introduction of impurity energy levels within the bandgap. Specifically, a clear correlation emerges between the organic electronegativity and E_g of VP-molecular heterojunctions. As illustrated in Figure 4k, VP-COOH and VP-NO₂, which are modified with molecules of lower electronegativity, exhibit larger E_g values compared to VP-TPA and VP-N(Et)₂, which are modified with molecules of more negative electronegativity. This

observation aligns with experimental data in Figure 3g, where donor molecules tend to narrow the bandgap more than acceptor molecules. Both the electronegativity-dependent charge-transfer dynamics and the associated bandgap modulations provide compelling evidence that the electronic properties of VP-molecular heterojunctions can be precisely manipulated at the molecular level.

To elucidate the electronegativity-dependent grating efficiency differences, we calculated the formation energy (E_f) of the

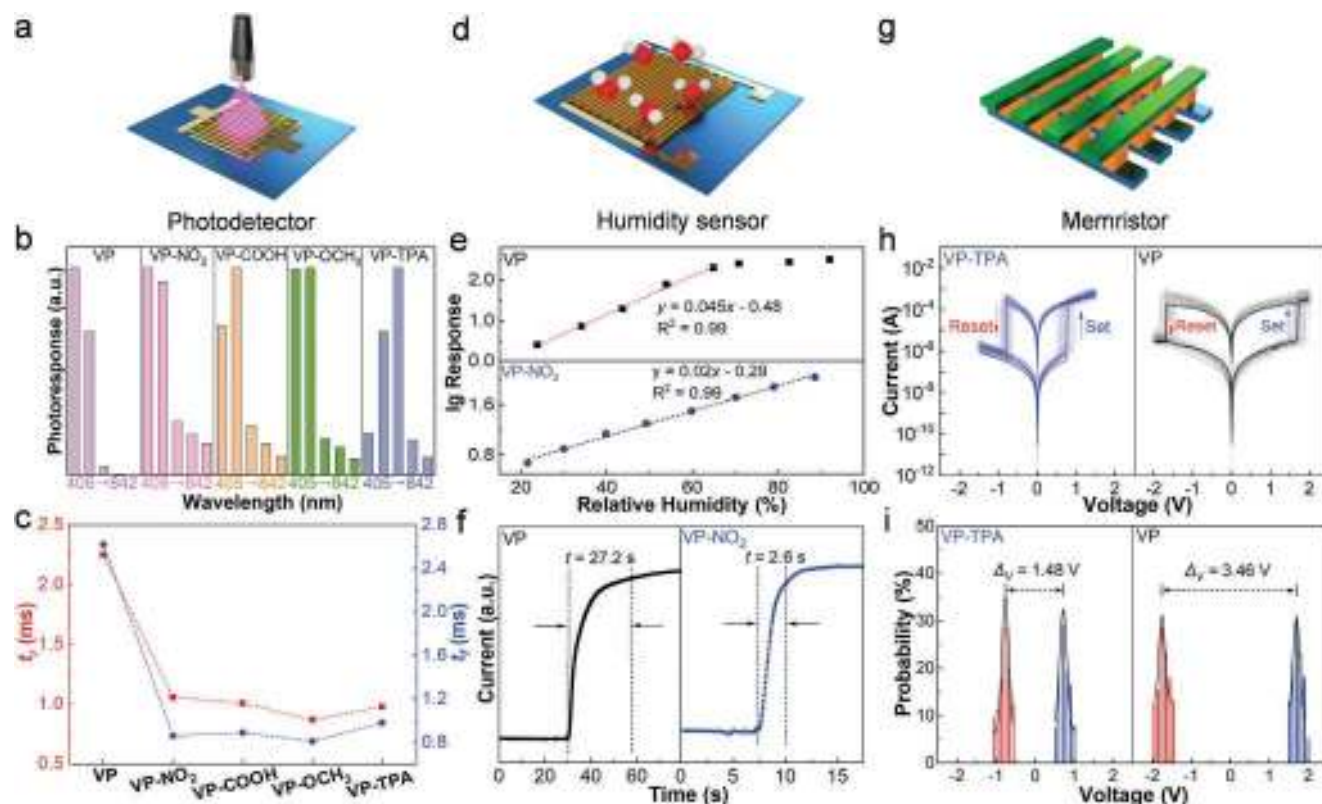


Figure 5. Demonstrations of VP-molecular heterojunction with improved device performance. a) Schematic illustration of the fabricated photodetector. b) Normalized broadband photoresponse comparison of VP, VP-NO₂, VP-COOH, VP-OCH₃ and VP-TPA-based photodetectors under 405–842 nm light illumination. c) Trend of response time at $\lambda = 505$ nm, with t_r and t_f representing rise and fall time, respectively. d) Schematic illustration of the fabricated humidity sensor. e) Humidity response comparison of VP and VP-NO₂-based humidity sensors as a function of the relative humidity from 20% to 90%. Dotted lines represent linear fits. f) Normalized response time of VP and VP-NO₂-based humidity sensors at 40% relative humidity. g) Schematic illustration of the fabricated memristor array. h, i) Comparison of I - V characteristics h) and the corresponding switching voltage distribution i) of VP-TPA and VP-based memristors.

eight VP-molecular heterojunctions, which reflects the difficulty of the chemical reaction. According to the reaction mechanism, E_f for the VP-molecular heterojunction is defined by the equation $E_f = E_{\text{VP-molecular heterojunction}} + E_{\text{nitrogen}} - E_{\text{VP}} - E_{\text{Organic molecule}}$, where $E_{\text{VP-molecular heterojunction}}$, E_{nitrogen} , E_{VP} and $E_{\text{Organic molecule}}$ represent the energies of VP-molecular heterojunction, nitrogen, VP and organic molecule, respectively. As shown in Figure 4l, the E_f of four VP-Donor heterojunctions generally falls within the range of 0 to -1 eV, whereas the E_f values of four VP-Acceptor heterojunctions range from -1 to -3 eV. The larger negative E_f values indicate the corresponding structures are more thermodynamically stable. Consequently, VP-Acceptor heterojunctions exhibit higher grafting efficiency than VP-Donor counterparts, consistent with the experiment results.

2.4. Demonstrations of VP-Molecular Heterojunctions in Typical Application Domains

The ability to precisely manipulate charge-transfer interactions and finely tune band structures endows VP-molecular heterojunctions with immense potential for practical applications.

Next, we focus on the demonstrations of VP-molecular heterojunctions in various application domains, with a particular emphasis on the performance modulation introduced by the organic molecules. Specifically, given the outstanding photosensitive properties of VP, the unique surfaces grafted by organic groups, and the improved charge-transfer dynamics, we explore three typical application fields: photodetectors, humidity sensors, and memristors, which represent some of the most promising research directions in the field of 2D materials.

As depicted in Figure 5a, photodetectors based on pristine VP and four representative VP-molecular heterojunctions were fabricated to reveal the photosensitivity differences induced by functionalization. Initially, the electrical conductivities of these five samples were evaluated using DC linear sweeps, as described in Note S8 (Supporting Information). Pristine VP exhibited relatively low conductivity, while the VP-molecular heterojunctions demonstrated significant improvements, indicating that organic functionalization effectively enhances VP's electrical properties. Next, we investigate the photoconductivity by illuminating the devices with laser beams. Considering the distinctive E_g variations, the broadband response capacity was characterized across the range of 405–842 nm. As shown in Figure 5b

and detailed in Note S8 (Supporting Information), the pristine VP-based photodetector exhibits a pronounced photoresponse when subjected to laser illumination at the 405 and 505 nm wavelengths. However, significant reductions in photoresponse are observed when the photon energies fall below the VP's E_g , particularly at 780 and 842 nm, which is consistent with our previous report.^[5] In contrast, the four VP-molecular heterojunction-based photodetectors demonstrate remarkably different photosensitive properties in terms of both maximum response wavelength and effective response band, notably broadening their sensitivity to the 660–842 nm range. Additionally, the response speeds of the five photodetectors were characterized. As shown in Figure 5c, the rise and fall times of the pristine VP-based photodetector are ≈ 2.3 and 2.6 ms, respectively. Following functionalization, the rise and fall times of the four VP-molecular heterojunction-based photodetectors decrease to below 1.1 and 1.0 ms, representing more than a twofold improvement. Moreover, the differences in responsivity (R) and specific detectivity (D^*) were calculated. Notably, the VP-OCH₃-based photodetector achieves a maximum R of 72.8 mA W⁻¹ and a D^* exceeding 3.5×10^{11} Jones, marking 14-fold and threefold improvements over the pristine VP counterpart, respectively. The high D^* of the VP-OCH₃ heterojunction also surpasses that of MoS₂ (2.1×10^{10}),^[46] PdPSe (2×10^{10}),^[47] SiAs (1×10^{10}),^[48] SnS (1.09×10^9),^[49] highlighting its considerable promise for dim-light detection applications. The observed enhancement in the photoresponse properties of VP-molecular heterojunctions is attributed to the reduced E_g and the enhanced charge-transfer effects, which deserve further detailed investigation in future studies.

Figure 5d illustrates the schematic diagram of the prepared humidity sensor, and the dynamic response properties of the VP-based and VP-NO₂-based sensors were evaluated using the humidity-sensing platform. As depicted in Figure 5e and detailed in Note S9 (Supporting Information), when the relative humidity (RH) increases from 20% to 90%, the VP-based humidity sensor demonstrates a linear response within the range of 20–60% RH, but this response gradually saturates as RH continues to rise. The saturation at higher RH levels is likely due to a balance between excessive adsorption of water molecules and disturbances in carrier transport caused by partial penetration. In contrast, the VP-NO₂-based humidity sensor exhibits a linear response across the entire RH range, which is essential for accurate RH quantification. The extended linear response range of the VP-NO₂ can be attributed to the excellent protective effect of the grafted nitrobenzene molecules, which prevent excessive water molecules from permeating into the host VP, thus maintaining a linear response even at high RH levels. Additionally, the transient response time of both humidity sensors was characterized. As shown in Figure 5f, the response time of the VP-NO₂-based humidity sensor is 2.6 s, which is more than ten times faster than that of the VP-based counterpart. We also compare the key performance of the VP-NO₂-based humidity sensor with other 2D material-based counterparts reported in the literature, highlighting its overall advantages for practical applications (Note S11, Supporting Information). This substantial improvement can be attributed to the enhanced hydrogen bonding interactions between water molecules and the nitro functional group in nitrobenzene, which facilitate a faster absorption rate and more efficient

charge transfer, thereby significantly improving the response speed.

Another significant research avenue in the realm of 2D materials is the development of 2D material-based memristors, which are considered crucial components for next-generation integrated electronics.^[50] We next investigate the application prospects of VP and VP-molecular heterojunction in the memristor field. Figure 5g depicts the schematic diagram of the fabricated memristor array, and the corresponding resistive switching properties of the VP-TPA-based and VP-based memristors are characterized. As shown in Figure 5h, VP-TPA-based and VP-based memristors exhibit nonvolatile, bipolar memristive properties under DC voltage excitation. Initially, both memristors are in a high resistive state (HRS) until a positive voltage is applied, switching the devices to a low resistive state (LRS), commonly referred to as the Set process. Conversely, the LRS switches back to HRS under negative voltages in the Reset process. The VP-TPA-based memristor demonstrates a significantly narrower switching bias window compared to the VP-based counterpart, with notably smaller switching threshold voltages, V_{set} and V_{reset} . Figure 5i provides a statistical comparison of V_{set} and V_{reset} for the two types of memristors. The average V_{set} and V_{reset} values for the VP-TPA-based memristor are 0.72 and -0.76 V, respectively, whereas, for the VP-based memristor, these values are 1.71 and -1.75 V. Notably, the difference (ΔV) in the V_{set} and V_{reset} for the VP-TPA-based memristor is less than half that of its VP-based counterpart. To provide a broader perspective, the switching threshold voltage of VP-TPA also surpasses that of *h*-BN, Bi₂O₃Se, and various TMDs-based memristors with similar device architectures (1–3 V).^[51–54] Our analysis of the resistive switching mechanism reveals that charge transfer plays a key role in resistive state conversions. In both VP and VP-TPA, electrons can be electrically excited from the valence band to the conduction band, enabling the devices to switch from HRS to LRS. Due to the narrower E_g and enhanced charge-transfer interactions of VP-TPA, smaller threshold voltages are required to achieve state conversion, which is advantageous for reducing power consumption in practical applications.

3. Conclusion

In summary, we have successfully developed a refined atomic-molecular heterojunction strategy, distinguished by robust and substantial covalent bonds between grafted organic molecules and 2D materials. By meticulously tuning the electronegativity of the grafted organic molecules, the degree of covalent functionalization, along with the band structures and charge-transfer properties of the synthesized VP-molecular heterojunctions, can be precisely controlled. This innovative approach opens a promising avenue for fine-tuning the electronic structure of VP at the molecular level, offering a versatile method for creating customizable 2D atomic-molecular heterojunctions aimed at expanding the tunability of 2D materials and enhancing device performance. In the future, atomic-molecular heterojunction technology could serve as a versatile platform to harness the synergistic effects of diverse material systems across a wide range of applications, heralding a new era of multifunctional 2D material technologies.

4. Experimental Section

Preparation of Few-layer VPNSs: Bulk VP crystals were synthesized via chemical vapor transport and ground into microcrystalline powder using an agate mortar. The resulting VP microcrystals were dispersed in ethanol and subjected to a custom-designed tip ultrasonication system to initiate the liquid phase exfoliation process. Following this, the VP samples underwent standard lithium intercalation to further reduce their thickness, yielding VPNSs with $\approx 3\text{--}4$ layers.

Covalent Functionalization of VP-Molecular Heterojunctions: The aryl diazonium functionalization method was employed to covalently modify the as-prepared few-layer VPNSs. Diazonium salts of nitrobenzene and methoxybenzene were purchased from Sigma-Aldrich, while diazonium salts of triphenylamine, N,N-Diethylaniline, N,N-Dimethylaniline, benzoic acid, trifluorotoluene, and fluorobenzene were synthesized through the diazotization of their respective aromatic amines. These diazonium salts were dissolved in acetonitrile and added dropwise to the VP nanosheet dispersions, with tetrabutylammonium hexafluorophosphate serving as the electrolyte. The reactions were conducted overnight in the dark to ensure complete functionalization. The resulting samples were thoroughly washed with acetonitrile to remove any unreacted diazonium salts and then freeze-dried to obtain the VP-molecular heterojunction powders.

Characterizations: Morphology characterizations were conducted using TEM (Talos F200X G2). The thickness of the VPNSs was measured using AFM (FastScan Bio). Raman and PL spectra were acquired with a spectrometer (Renishaw inVia Qontor). FTIR and UV–Vis–NIR measurements were recorded using a Micro-infrared spectrometer (iN10 MX) and a spectrophotometer (Lamda 950), respectively. To quantize the covalent modification efficiency, element analysis was performed using an elemental analyzer (Vario EL Cube).

Device Fabrications and Measurements: VP-based and VP-molecular heterojunction-based photodetectors and humidity sensors were fabricated using the drop-casting technique to disperse the samples onto interdigitated electrodes (IDEs) with a film thickness of ≈ 120 nm. These IDEs comprise 50 pairs of Au strips with a small channel length ($L = 2\text{ }\mu\text{m}$) and a total conduction channel width ($W = 4950\text{ }\mu\text{m}$). For VP-based and VP-NO₂ heterojunction-based memristor crossbars, indium tin oxide (ITO) was used as the bottom electrode, and gold (Au) as the top electrode, both deposited via magnetron sputtering. In this configuration, the thicknesses of the ITO, VP functional layer, and Au are 80, 50, and 100 nm, respectively. The photoresponse was characterized using five laser sources (405, 505, 660, 780, and 842 nm), with laser power determined by a dynamometer (Thorlabs). Humidity sensing tests were conducted under precisely controlled relative humidity conditions ranging from 20% to 90%, achieved by adjusting the mass flow ratio of dry to humidified gas, and calibrated using a high-precision hygrometer. Memristive characterizations were performed using a 4200A semiconductor parameter analyzer (Keithley).

Computational Methods: In this study, calculations were conducted using density functional theory (DFT) with the Vienna Ab initio Simulation Package (VASP).^[55] The electron exchange and correlation were described using the generalized gradient approximation (GGA), as formulated by Perdew, Burke, and Ernzerhof (PBE).^[56] The DFT-D3 method was applied for van der Waals corrections in all calculations.^[57,58] A plane-wave basis set was utilized with a kinetic cutoff energy of 450 eV. A 20 Å vacuum layer was created to eliminate interactions with adjacent cells. The Monkhorst–Pack scheme K points grid sampling was set as $2 \times 4 \times 1$ for the irreducible Brillouin zone.^[59] For the computation of a single molecule, a $20 \times 20 \times 20$ Å³ unit cell was used with $2 \times 2 \times 2$ Monkhorst–Pack meshes. Recognizing that the GGA function tends to underestimate bandgaps, the HSE06 hybrid functional was employed to calculate the band structures.^[60,61] A K points mesh of $2 \times 3 \times 1$ was used for the geometric optimization of 4 layers VP-NO₂. The projected area of molecular clusters on both the upper and lower surfaces was determined using a combination of DBSCAN clustering and Convex Hull analysis. DBSCAN clustering (with an epsilon of 1.5 and a minimum sample size of 1) was applied to the molecule coordinates to identify distinct clusters. For each cluster, the Convex Hull was computed to determine the projected area.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

W.C., A.C., X.L., and F.S. contributed equally to this work. W.C. and G.L. conceived the idea. J.Z. synthesized the bulk VP crystals. W.C. synthesized the atomic-molecular heterojunctions, fabricated the devices, and conducted the optoelectronic and memristive measurements. W.C., X.L., and Z.Y. prepared the humidity sensors and carried out the humidity sensing tests. W.C., H.X., and G.P. conducted electrical and long-term stability tests of the VP-organic heterojunctions and analyzed the data. A.C., F.S., J.Z., and J.L. carried out DFT calculations and related theoretical analyses. W.C., A.C., and G.L. co-wrote the paper. All the authors discussed the results and commented on the manuscript.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

atomic-molecular heterojunction, band structure regulation, charge-transfer, covalent bond, organic electronegativity, violet phosphorus

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